

REVA UNIVERSITY, BENGALURU

Faculty Development Programme (FDP)

Federated Learning for Secure and Distributed AI Systems



Date: 22/09/2026



05:00 pm to 08:00 pm | 3 Hours



Day 1: Foundations of Artificial Intelligence and Distributed Learning



Programme at a Glance

5

Days

3

Hours per Day

15

Total Contact Hours



Date

22/09/2026



Daily Timing

05:00 pm to 08:00 pm



Host Institution

REVA University, Bengaluru

Programme Objectives



Strengthen AI Foundations

Revisit the core ideas of AI, machine learning and deep learning that support advanced topics.



Understand Distributed Learning

Compare centralized, distributed and federated approaches and their trade-offs.



Learn Privacy-Preserving AI

Train useful models without moving sensitive data, in line with data-protection needs.



Gain Hands-on Exposure

Explore FL frameworks and use cases that can be used in teaching, research and projects.

Day 1 Agenda

1

05:00 – 05:40 pm

**AI, ML, Deep
Learning & Intelligent
Systems**

Concepts · Real examples

2

05:40 – 06:20 pm

**Centralized vs
Distributed Learning**

Models · Challenges

3

06:20 – 07:10 pm

Federated Learning

Architecture · Workflow ·
Applications

4

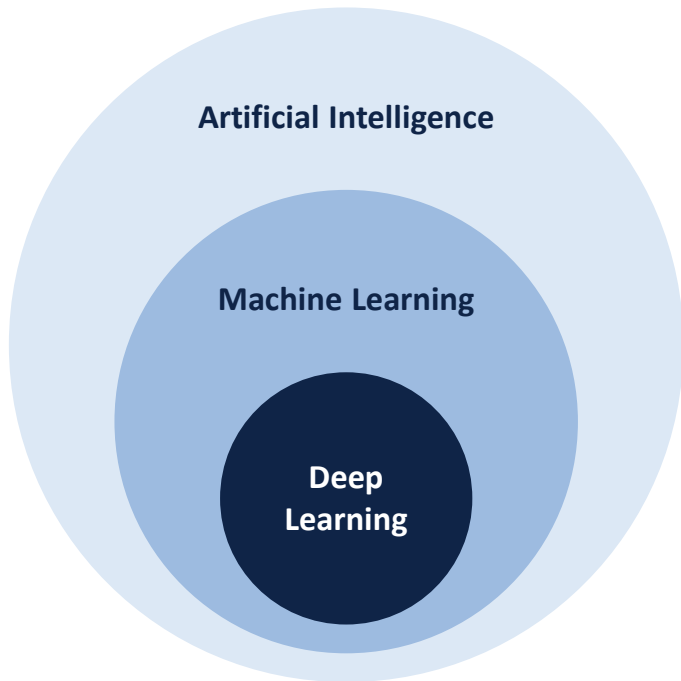
07:10 – 08:00 pm

Demonstration

FL environments · Use
cases

Proposed session flow for 3 hours (05:00 pm to 08:00 pm). Timings are flexible.

AI, Machine Learning and Deep Learning



Artificial Intelligence

Machines that imitate human abilities such as reasoning, planning and decision-making.

Example: route planning in Google Maps.

Machine Learning

Algorithms that learn patterns from data instead of following fixed rules.

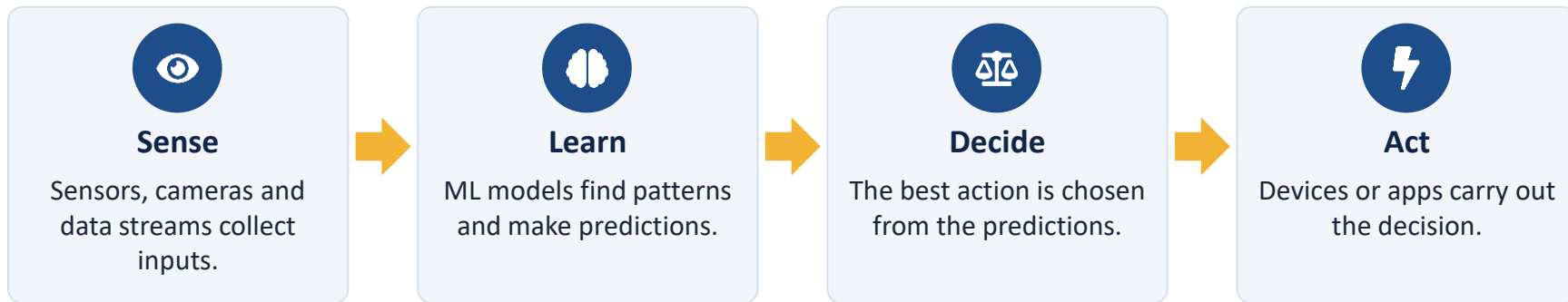
Example: a spam filter learning from labelled emails.

Deep Learning

Multi-layer neural networks that learn features automatically from large data.

Example: face unlock and voice typing on phones.

How an Intelligent System Works



REAL-TIME EXAMPLE · Adaptive traffic signals in a busy city

Cameras count vehicles waiting at each junction.

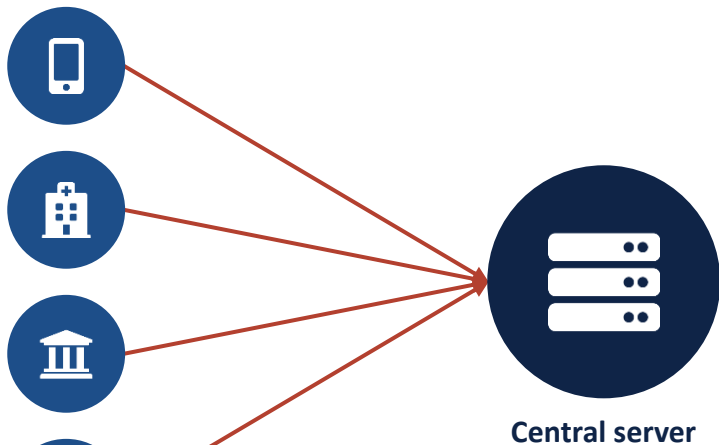
A model predicts the queue that will build in the next few minutes.

Green time is extended for the busiest road.

Signals change, and the system keeps improving with new data.

Centralized Learning and Its Limits

How it works



Raw data is copied to one place, then the model is trained there.

Challenges



Privacy risk

Sensitive data leaves the device or hospital.



Legal restrictions

Laws such as GDPR and India's DPDP Act 2023 limit data transfer.



Bandwidth and storage cost

Moving huge datasets is slow and expensive.



Single point of failure

One breach or outage exposes or halts everything.

Distributed Learning: Models and Challenges



Data Parallelism

The same model runs on every node, each on a different slice of data. Results are merged.

Example: training an image classifier on 8 GPUs, each handling different batches.



Model Parallelism

The model is split across nodes because it is too large for a single machine.

Example: a very large language model divided layer by layer across many GPUs.

Key challenges



Communication overhead

Frequent syncing of models slows training.



Data heterogeneity

Each node sees a different (non-IID) slice of data.



Device heterogeneity

Nodes differ in speed, memory and battery.



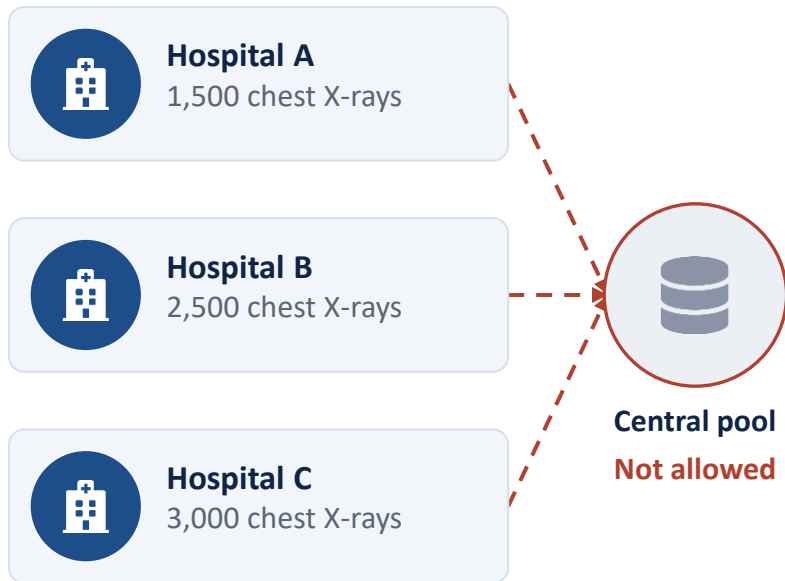
Reliability and security

Dropouts, slow nodes and malicious nodes.

Centralized vs Distributed vs Federated

Aspect	Centralized	Distributed	Federated
Where data lives	One central server	Shards across cluster nodes	Stays with each data owner
What is shared	Raw data	Data shards and gradients	Only model updates
Privacy	Low: data is pooled	Moderate: data still pooled	High by design
Typical use	Single-organisation ML	Large training in data centres	Phones, hospitals, banks, IoT
Main limitation	Privacy, law, bandwidth	Sync and communication cost	Non-IID data, device variety

The Hospital Data Dilemma



Scenario

Three hospitals want an AI model that detects lung disease from chest X-rays. Each holds only a few thousand images, which is not enough for an accurate model.

Roadblock

Pooling all images on one server is restricted by patient-privacy laws (India's DPDP Act 2023, HIPAA, GDPR) and by hospital policy.

Can the model learn from all three hospitals without any image leaving its hospital?

That is exactly what Federated Learning does.

What is Federated Learning?

Federated Learning (FL) is a machine-learning approach in which many clients (phones, hospitals, banks, sensors) jointly train one shared model while their raw data stays where it is. Only model updates are sent to a coordinating server.

Introduced by Google researchers (McMahan et al., 2017) through the Federated Averaging (FedAvg) algorithm.

Traditional ML: bring the data to the model



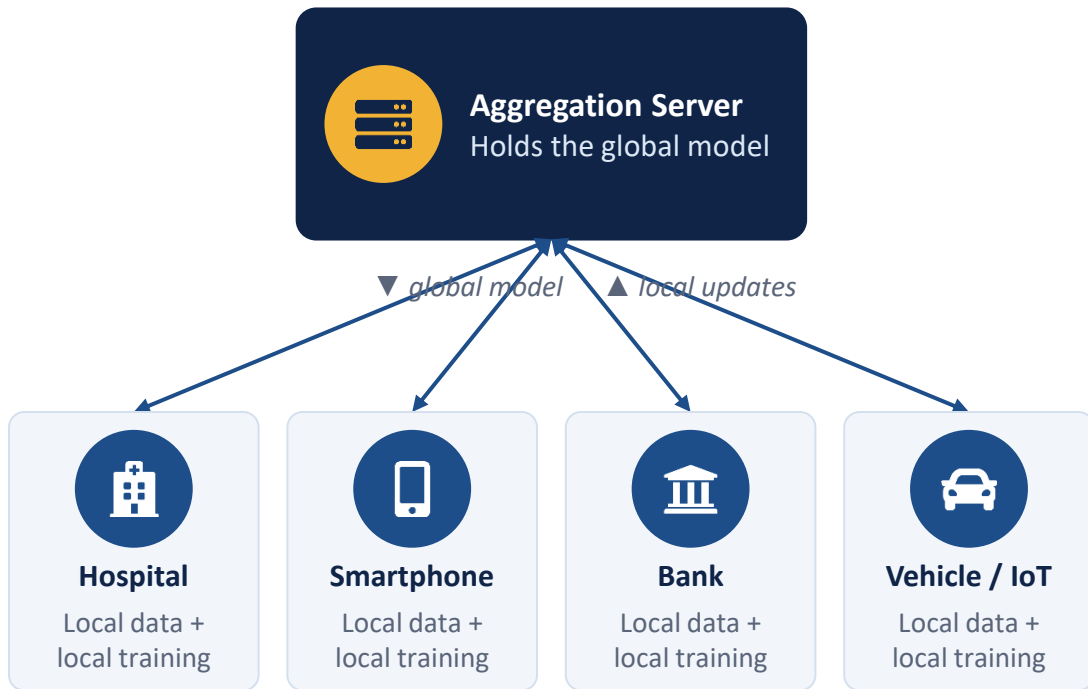
Raw data is uploaded to a central server, where a single model is trained.

Federated Learning: bring the model to the data



The model travels to each client. Only learned updates come back, never raw data.

Federated Learning Architecture



Key components

- 1 Server**
Coordinates rounds and aggregates updates.
- 2 Clients**
Hold private data and train the model locally.
- 3 Local data**
Never leaves the client.
- 4 Communication**
Only model updates travel, ideally encrypted.

Federated Learning Workflow

1

Initialize

The server creates the first version of the global model.

2

Select and Send

The server picks a group of clients and sends them the current model.

3

Train Locally

Each client trains the model on its own data for a few epochs.

4

Upload Updates

Clients return only the updated weights, never the data.

5

Aggregate

The server combines the updates (for example FedAvg) into a new global model.

6

Repeat

Rounds continue until the model reaches the desired accuracy.

Aggregation with FedAvg: A Simple Calculation

Client	Records (n_k)	Share (n_k / n)	Local weight (w_k)	Contribution
Hospital A	1,000	10%	0.20	$0.10 \times 0.20 = 0.020$
Hospital B	3,000	30%	0.40	$0.30 \times 0.40 = 0.120$
Hospital C	6,000	60%	0.50	$0.60 \times 0.50 = 0.300$
Total	10,000	100%		Global weight = 0.440

$$w(\text{global}) = \sum (n_k \div n) \times w_k$$

n_k = records at client k , n = total records, w_k = client k 's local weight

Why weighted?

A plain average gives 0.37. FedAvg gives 0.44 because Hospital C holds 60% of the data, so it influences the global model more.

Types of Federated Learning



BY PARTICIPANTS

Cross-Device FL

Millions of phones or IoT devices, each with little data and frequent dropouts.

Example: Gboard keyboard.



BY PARTICIPANTS

Cross-Silo FL

A few organisations with large, reliable datasets and stable connections.

Example: hospitals or banks training together.



BY DATA SPLIT

Horizontal FL

Same features, different users or samples.

Example: two hospitals with the same patient record format.



BY DATA SPLIT

Vertical FL

Same users, different features.

Example: a bank and an online retailer serving the same customers.

Next-Word Prediction on Your Phone

1

Download

The phone gets the current prediction model, usually when idle, charging and on Wi-Fi.

2

Learn on device

As you type, the phone improves the model locally using your own typing.

3

Send an update

Only a small, encrypted model update is sent, never your messages.

4

Combine securely

The server averages updates from thousands of phones (secure aggregation).

5

Improve for all

The better model returns to everyone, with smarter suggestions for new words and phrases.



Why Federated?

- Messages, names and passwords stay on the phone.
- Millions of users improve one shared model.
- New slang and trending words are learned quickly.
- Battery and data use stay low.

Hospitals Predicting COVID-19 Outcomes



The problem

Each hospital had too little COVID-19 data alone, and patient records could not be shared across borders.



The FL approach

20 hospitals trained the model on their own data. Only model updates were combined centrally.



The outcome

A single model that predicts a patient's oxygen needs, and works better across hospitals than models trained at one site.

ILLUSTRATIVE USE CASE



Banks and Fraud Detection

Banks A, B and C keep customer transactions private.

A shared fraud model learns patterns seen at all three banks.

New fraud tricks found at one bank protect the others faster.

Source: Dayan et al., Nature Medicine (2021), EXAM study.

Applications Across Domains



Healthcare

Medical imaging and disease prediction across hospitals.



Finance

Fraud detection and credit scoring across banks.



Mobile and Consumer

Keyboards, voice assistants and recommendations.



Autonomous Vehicles

Learning from many cars' driving data without uploading it.



Smart Cities and IoT

Traffic, energy grids and sensor networks.



Industry 4.0

Predictive maintenance across factories and plants.

Benefits and Challenges of FL



Benefits

- **Privacy by design:** raw data never leaves its owner.
- **Easier compliance:** supports data-protection requirements.
- **Lower bandwidth:** only small updates travel.
- **Richer learning:** uses diverse real-world data.
- **Personalization:** models adapt to local users.



Challenges

- **Non-IID data:** each client's data looks different.
- **Communication cost:** many rounds over slow networks.
- **Device variability:** speed, memory and battery differ.
- **Security threats:** model poisoning, inference attacks.
- **Fairness:** clients with more data may dominate.

Note: FL lowers privacy risk but is not automatically private. It is combined with secure aggregation and differential privacy.

Federated Learning Frameworks



Flower (flwr)

Framework-agnostic and beginner-friendly. Works with PyTorch, TensorFlow and more.



TensorFlow Federated

Google's library for FL research and simulation on TensorFlow.



PySyft (OpenMined)

Privacy-preserving ML and remote data science on private data.



NVIDIA FLARE

Production-oriented FL, widely used in healthcare and enterprise.



FedML

Open-source library and platform for cross-device and cross-silo FL.

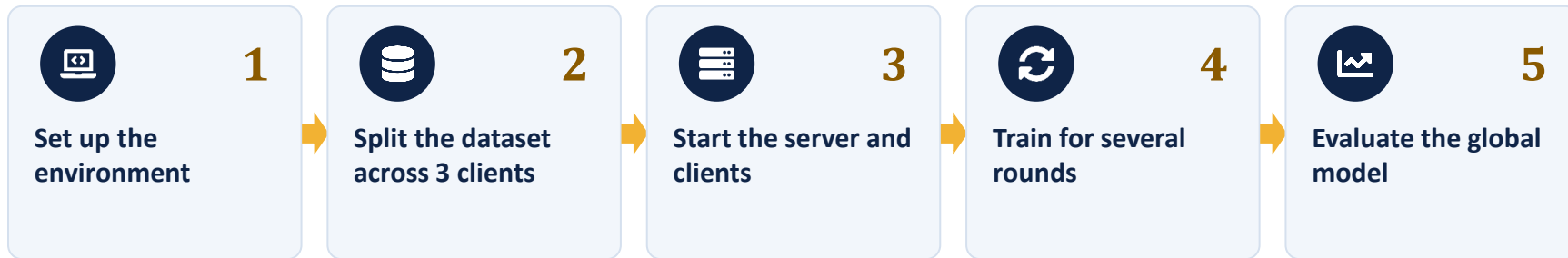


FATE

Industrial-grade FL platform from WeBank, strong in finance.

Suggested for today's demo: Flower with PyTorch or TensorFlow, simulated on a laptop.

Demo: Federated Training with 3 Clients



What we will observe

- Accuracy improving round after round
- Global model vs each local model
- Effect of uneven (non-IID) data splits
- Raw data never leaves a client

Use cases we can try

- Handwritten digits (MNIST) across 3 "banks"
- Chest X-ray classification across 3 "hospitals"
- Keyboard-style text prediction across "phones"

Key Takeaways



**Bring the model to
the data, not the data
to the model.**

1

AI contains ML, which contains deep learning. Intelligent systems sense, learn, decide and act.

2

Centralized learning struggles with privacy, regulation and bandwidth.

3

Distributed learning scales training but adds communication and heterogeneity challenges.

4

Federated Learning trains a shared model where the data lives. Only updates travel.

5

Real systems such as Gboard and hospital networks already use FL.

Thank You

Questions and Discussion

FDP on Federated Learning for Secure and Distributed AI Systems

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